

CHAPTER 3

3-1. (a) $E^*(s) = \sum_{n=0}^{\infty} e(nT) e^{-nTs}$ (b) $E(z) = \sum_{n=0}^{\infty} e(nT) z^{-n}$
 (c) $E^*(s) = E(z) \big|_{z=e^{sT}}$

- 3-2. (a) 1. No frequencies in $e(t)$ greater than $\omega_s/2$.
 2. An ideal low-pass filter follows the sampler.
 (b) None
 (c) The signal can be recovered to a sufficient degree of accuracy.

3-3.
$$\frac{E(s)}{T} \xrightarrow{E^*(s)} \boxed{G_{ho}(s)} \xrightarrow{\bar{E}(s)}$$

$$E_1(s) \longrightarrow \bar{E}_1(s) \quad E_2(s) \longrightarrow \bar{E}_2(s)$$

Let $E = E_1 + E_2$
 $\therefore E^* = (E_1 + E_2)^* = E_1^* + E_2^*$
 $\therefore \bar{E} = G_{ho}(E_1^* + E_2^*) = G_{ho}E_1^* + G_{ho}E_2^*$

3-4. $E^*(s) = \sum \text{residues of } E(\lambda) \frac{1}{1 - e^{-T(s-\lambda)}}$

(a)
$$E^*(s) = \left. \frac{20}{(s+5)(1-e^{-T(s-\lambda)})} \right|_{\lambda=-2} + \left. \frac{20}{(s+2)(1-e^{-T(s-\lambda)})} \right|_{\lambda=-5}$$

$$= \frac{20/3}{1-e^{-T(s+2)}} + \frac{-20/3}{1-e^{-T(s+5)}}$$

(b)
$$E^*(s) = \left. \frac{5}{(s+1)(1-e^{-T(s-\lambda)})} \right|_{\lambda=0} + \left. \frac{5}{s(1-e^{-T(s-\lambda)})} \right|_{\lambda=-1}$$

$$= \frac{5}{1-e^{-Ts}} - \frac{5}{1-e^{-T(s+1)}}$$

$$3-4.(c) E^*(s) = \frac{\lambda+2}{(\lambda+1)(1-e^{-T(s-\lambda)})} \Big|_{\lambda=0} + \frac{\lambda+2}{\lambda(1-e^{-T(s-\lambda)})} \Big|_{\lambda=-1} = \frac{2}{1-e^{-Ts}} - \frac{1}{1-e^{-T(s+1)}}$$

$$\begin{aligned} (d) \text{ (residue)}_{\lambda=0} &= \frac{d}{d\lambda} \left[\frac{\lambda+2}{(\lambda+1)(1-e^{-T(s-\lambda)})} \right]_{\lambda=0} \\ &= \frac{(\lambda+1)(1-e^{-T(s-\lambda)}) - (\lambda+2)(\lambda+1)(-Te^{-T(s-\lambda)}) - (\lambda+2)(1-e^{-T(s-\lambda)})}{(\lambda+1)^2(1-e^{-T(s-\lambda)})^2} \Big|_{\lambda=0} \\ &= \frac{-(1-e^{-Ts}) + 2Te^{-Ts}}{(1-e^{-Ts})^2} \end{aligned}$$

$$\text{(residue)}_{\lambda=-1} = \frac{\lambda+2}{\lambda^2(1-e^{-T(s-\lambda)})} \Big|_{\lambda=-1} = \frac{1}{1-e^{-T(s+1)}}$$

$$\therefore E^*(s) = \frac{-(1-e^{-Ts}) + 2Te^{-Ts}}{(1-e^{-Ts})^2} + \frac{1}{1-e^{-T(s+1)}}$$

$$(e) E^*(s) = \sum_{\text{residues}} \frac{\lambda^2 + 5\lambda + 6}{\lambda(\lambda+4)(\lambda+5)(1-e^{-T(s-\lambda)})} = \frac{3/10}{1-e^{-Ts}} + \frac{-1/2}{1-e^{-T(s+4)}} + \frac{6/5}{1-e^{-T(s+5)}}$$

$$(f) s = -1 \pm j2$$

$$\begin{aligned} E^*(s) &= \frac{z}{(\lambda+1+j)(1-e^{-T(s-\lambda)})} \Big|_{\lambda=-1+j2} + \frac{z}{(\lambda+1-j2)(1-e^{-T(s-\lambda)})} \Big|_{\lambda=-1-j2} \\ &= \frac{-j/2}{1-e^{-T(s+1-j2)}} + \frac{j/2}{1-e^{-T(s+1+j2)}} \end{aligned}$$

$$\begin{aligned} 3-5.(a) E^*(s) &= 1 + e^{aT}e^{-Ts} + e^{2aT}e^{-2Ts} + \dots = 1 + e^{(a-s)T} + [e^{(a-s)T}]^2 + \dots \\ &= \frac{1}{1-e^{(a-s)T}} \end{aligned}$$

$$(b) e(t) = e^{a(t-2T)} u(t-2T)$$

$$\begin{aligned} E^*(s) &= e^{-2Ts} + e^{aT}e^{-3Ts} + e^{2aT}e^{-4Ts} + \dots \\ &= e^{-2Ts}(1 + e^{aT}e^{-Ts} + e^{2aT}e^{-2Ts} + \dots) = \frac{e^{-2Ts}}{1-e^{(a-s)T}} \end{aligned}$$

$$(c) \text{ From (b), } E^*(s) = \frac{e^{-2Ts}}{1-e^{(a-s)T}}$$

$$(d) E^*(s) = e^{aT/2}e^{-Ts} + e^{3aT/2}e^{-2Ts} + e^{5aT/2}e^{-3Ts} + \dots$$

$$3-5.(d) \quad E^*(s) = e^{aT/2} e^{-Ts} (1 + e^{aT} e^{-Ts} + e^{2aT} e^{-2Ts} + \dots) \\ = \frac{e^{aT/2} e^{-Ts}}{1 - e^{(a-s)T}}$$

$$3-6. (a) \quad E^*(s) = 1 + e^{-3T} e^{-Ts} + e^{-6T} e^{-2Ts} + \dots$$

$$(b) \quad E^*(s) = \frac{1}{1 - e^{-T(s+3)}}$$

$$(c) \quad E(s) = \frac{1}{s+3} \quad \therefore \text{from (3-11),}$$

$$E^*(s) = \frac{1}{2} + \frac{1}{T} \left[\frac{1}{s+3} + \frac{1}{s+3+j\omega_s} + \frac{1}{s+3+j2\omega_s} + \dots \right. \\ \left. + \frac{1}{s+3-j\omega_s} + \frac{1}{s+3-j2\omega_s} + \dots \right]$$

$$3-7. \quad E^*(s) = e(0) + e(T) e^{-Ts} + e(2T) e^{-2Ts} + \dots = [e(kT)]^*$$

$$\text{Let: } e_1(t) = e(t - kT) u(t - kT)$$

$$\therefore E_1^*(s) = e(0) e^{-kTs} + e(T) e^{-(k+1)Ts} + \dots$$

$$= e^{-kTs} [e(0) + e(T) e^{-Ts} + \dots] = e^{-kTs} E^*(s)$$

$$\therefore \text{from (3-14),}$$

$$E_1^*(s) = e^{-kTs} \left[\sum_{\substack{\text{poles} \\ \text{of } E(s)}} \text{residues of } \frac{E(s)}{1 - e^{-T(s-\lambda)}} \right]$$

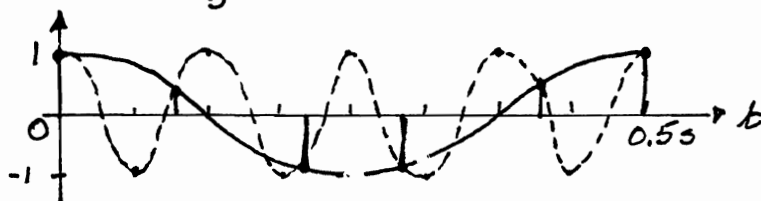
$$3-8. \quad E(s) = \frac{1 - e^{-Ts}}{s(s+1)} \quad ; \quad \text{define } E_1(s) = \frac{1}{s(s+1)}$$

$$\therefore E_1^*(s) = \frac{1}{(s+1)(1 - e^{-T(s-\lambda)})} \Big|_{\lambda=0} + \frac{1}{\lambda(1 - e^{-T(s-\lambda)})} \Big|_{\lambda=-1} \\ = \frac{1}{1 - e^{-Ts}} - \frac{1}{1 - e^{-T(s+1)}}$$

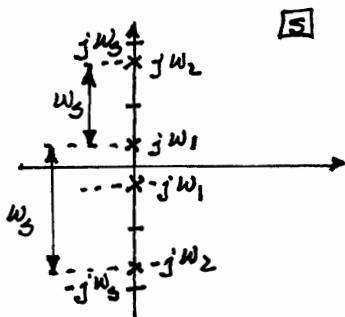
$$E(s) = E_1(s) - E_1(s) e^{-Ts} \quad ; \quad \therefore E^*(s) = E_1^*(s) - E_1^*(s) e^{-Ts}$$

$$= \left[\frac{1}{1 - e^{-Ts}} - \frac{1}{1 - e^{-T(s+1)}} \right] (1 - e^{-Ts}) = \frac{e^{-Ts} (1 - e^{-T})}{1 - e^{-T(s+1)}}$$

3-9. (a) (i) $\omega_1 = 4\pi$, $\omega_3 = \frac{2\pi}{T} = 20\pi$ (ii) $\omega_2 = 16\pi = \frac{4\omega_3}{5} = 4\omega_1$
 $\omega_1 = \frac{\omega_3}{5}$, $T_1 = 0.5$



$$\begin{aligned} \cos 72^\circ &= 0.309 & \cos(4 \times 72^\circ) &= 0.309 \\ \cos 144^\circ &= -0.809 & \cos(4 \times 144^\circ) &= -0.809 \\ \cos 216^\circ &= -0.809 & \cos(4 \times 216^\circ) &= -0.809 \\ \cos 288^\circ &= 0.309 & \cos(4 \times 288^\circ) &= 0.309 \end{aligned}$$



$$\begin{aligned} \omega_1 &= \omega_3 - \omega_2 \\ -\omega_1 &= -\omega_3 + \omega_2 \end{aligned}$$

(b) $\omega_3 = \omega_3 + \omega_1 = 20\pi + 4\pi = 24\pi \Rightarrow e_3(t) = \cos(24\pi t)$

3-10. $E(s) = \frac{s+2}{s(s+1)}$, $\therefore$ poles: $s=0, -1$, zero: $s=-2$

From Problem 3-4(c): $E^*(s) = \frac{2}{1-e^{-Ts}} - \frac{1}{1-e^{-T(s+1)}}$
 $= \frac{1 + e^{-Ts} - 2e^{-Ts}e^{-T}}{(1-e^{-Ts})(1-e^{-T(s+1)})}$

zeros: $1 + e^{-Ts}(1 - 2e^{-T}) = 0 \Rightarrow e^{Ts} = 2e^{-T} - 1$

$\therefore e^{\sigma T} = |2e^{-T} - 1|$

$e^{j\omega T} = 1/\omega T = \begin{cases} 1, & 2e^{-T} - 1 > 0 \Rightarrow \omega T = \pm n\pi, n=0, 2, 4, \dots \\ -1, & 2e^{-T} - 1 < 0 \Rightarrow \omega T = \pm n\pi, n=1, 3, 5, \dots \end{cases}$

$\therefore \omega = \pm n \frac{\pi}{T} = \pm n \frac{\omega_3}{2}$

$\therefore$ zeros: $\sigma \pm jn \frac{\omega_3}{2} \begin{cases} n=0, 2, 4 & \text{for } (1) \\ n=1, 3, 5 & \text{for } (-1) \end{cases}$

poles: $s = \pm jn\omega_3, -1 \pm jn\omega_3; n=0, 1, 2, \dots$

3-11. Consider $E_1(s) = \frac{1}{s^2(s+1)}$; Then $E^*(s) = E_1^*(s) [2(1-e^{-Ts})^2]$

$$(\text{residue})|_{\lambda=0} = \frac{d}{d\lambda} \left[\frac{1}{(\lambda+1)(1-e^{-T(s-\lambda)})} \right]_{\lambda=0}$$

$$= \frac{-(1-e^{-T(s-\lambda)}) - (-T e^{-T(s-\lambda)}) (\lambda+1)}{(\lambda+1)^2 (1-e^{-T(s-\lambda)})^2} \Big|_{\lambda=0} = \frac{-1 + e^{-Ts} + T e^{-Ts}}{(1-e^{-Ts})^2}$$

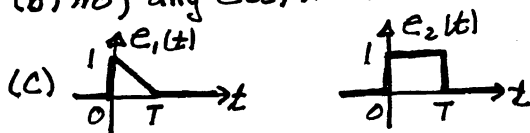
$$(\text{residue})|_{\lambda=-1} = \left[\frac{1}{\lambda^2 (1-e^{-T(s-\lambda)})} \right]_{\lambda=-1} = \frac{1}{1-e^{-T(s+1)}}$$

$$\therefore E_1^*(s) = \frac{1}{1-e^{-T(s+1)}} + \frac{-1 + e^{-Ts} + T e^{-Ts}}{(1-e^{-Ts})^2}$$

$$\therefore E^*(s) = \frac{1-e^{-Ts} - 1 + e^{-Ts} + T e^{-Ts}}{(1-e^{-T(s+1)})(1-e^{-Ts})^2} [2(1-e^{-Ts})^2] = \frac{2T e^{-Ts}}{1-e^{-T(s+1)}}$$

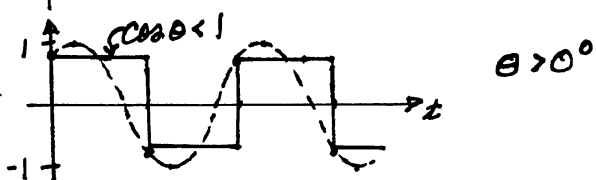
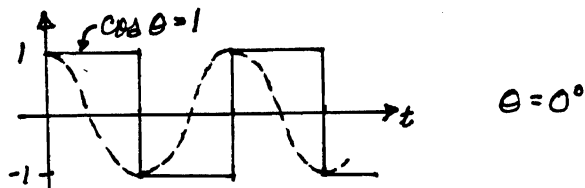
3-12. (a) $e(kT) = \begin{cases} 1, & k=0 \\ 0, & k \geq 1 \end{cases}$

(b) No, any $e(t)$ with $e(\omega) = 1$ and $e(kT) = 0, k \geq 1$



(d) $e_1(t) = (1 - \frac{1}{T}t)[u(t) - u(t-T)]$; $e_2(t) = u(t) - u(t-T)$

3-13. $e(t) = \cos(\frac{\omega_s}{2}t + \theta)$



(b) Output is a square wave with amplitude equal to $\cos \theta$.

$$\therefore \bar{e}(t) = A_0 + A_1 \cos \frac{\omega_s}{2}t + B_1 \sin \frac{\omega_s}{2}t + \dots$$

$$A_1 = \underline{0}, \text{ since } \bar{e}(t) \text{ is odd.}$$

$$3-13.(b) \quad B_1 = \frac{2}{T} \int_0^T \bar{e}(t) \sin(\frac{\omega_s}{2} t) dt = \frac{4}{T} \int_0^{T/2} \cos \theta \sin(\omega_s/2 t) dt$$

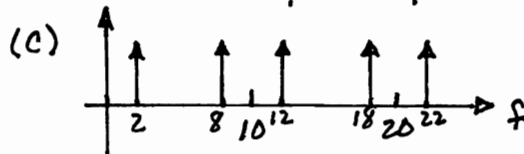
$$= \frac{4 \cos \theta}{T} \left[-\frac{\cos(\frac{\omega_s}{2} t)}{\omega_s/2} \right]_0^{T/2} = \frac{4}{\pi} \cos \theta, \text{ since } \omega_s = \frac{2\pi}{T}$$

$$\therefore 1^{st} \text{ harmonic} = \left(\frac{4}{\pi} \cos \theta \right) \sin\left(\frac{\omega_s}{2} t\right)$$

$$3-14. (a) f_s = 10 \text{ Hz}, f_1 = 2 \text{ Hz}.$$

$$(b) f_s = 10 \text{ Hz}, f_1 = 8 \text{ Hz}$$

	$\frac{a}{2}$	$\frac{b}{2}$
f_1	2	8
$f_s - f_1$	8	2
$f_s + f_1$	12	18
$2f_s - f_1$	18	12
$2f_s + f_1$	22	28
$\vdots$	$\vdots$	$\vdots$



$$f = 12, 18, 22, 28, 32, \dots$$

$$52, 58, 62, \dots$$

$$3-15. (a) \quad 4, 7 \text{ rad/s}$$

(b)	$\frac{4}{22 \pm 4 = \underline{18, 26}}$	$\frac{7}{22 \pm 7 = \underline{15, 29}}$
	$44 \pm 4 = \underline{40, 48}$	$44 - 7 = \underline{37}$

(c) Same as (b)

(d) Same as (b)

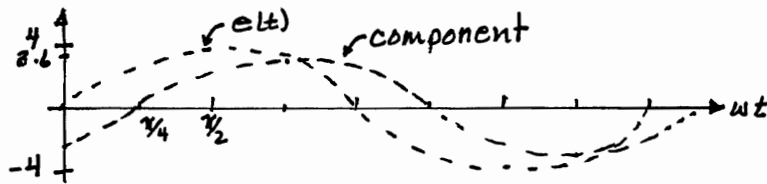
$$3-16. (a) \text{ Frequencies in } \bar{e}(t): \quad \underline{7}; 4 \pm 7 = \underline{-3, 10}; 8 \pm 7 = \underline{1, 15}; 12 \pm 7 = \underline{5, 19}$$



$$(b) \frac{1}{T} G_{ho}(j1) = \frac{\sin(\frac{\pi(1)}{4})}{\pi(1)/4} e^{-j\pi/4} = 0.900 \angle -45^\circ$$

$$\therefore \text{Component} = (4)(0.9) \sin(t - 45^\circ) = \underline{3.6 \sin(t - 45^\circ)}$$

3.16.(C)



$$(d) \frac{1}{T} G_{ho}(j7) = \frac{\sin(\pi 7/4)}{\pi 7/4} e^{-j\pi 7/4} = (-0.129)e^{-j315^\circ} = 0.129 \angle 135^\circ$$

$$\therefore \text{component} = 4(0.129) \sin(7t - 135^\circ) = 0.516 \sin(7t - 135^\circ)$$

$$\text{ratio} = \frac{3.6}{0.516} = 6.977 \quad (7)$$

3-17. (a) phase of 3. \theta. h. = $\pi \frac{\omega}{\omega_s} \Rightarrow \frac{180^\circ(10)}{\omega_s} = 10^\circ, \therefore \omega_s = 180 \text{ rad/s}$

$$T = 2\pi/\omega_s = 2\pi/180 = 0.0349 \text{ s}$$

(b) $\frac{180^\circ}{\omega_s} = 5^\circ \Rightarrow \omega_s = 360 \text{ rad/s} ; T = \frac{2\pi}{360} = 0.0175 \text{ s}$

(c) $\frac{180^\circ(10)}{\omega_s} = 20^\circ, \therefore \omega_s = 90 \text{ rad/s} ; T = 0.0698 \text{ s}$

3-18. (a) $\frac{1}{T} G_{ho}(j2) = \frac{\sin(\pi \frac{2\pi(2)})}{2\pi/12} e^{-j\pi/6} = \frac{0.50 \angle -30^\circ}{0.5236} = 0.9549 \angle -30^\circ$

(b) $\frac{1}{T} G_{ho}(j10) = \frac{\sin(\pi \frac{10\pi}{12})}{10\pi/12} e^{-j10\pi/12} = \frac{0.50 \angle -150^\circ}{2.618} = 0.1910 \angle -150^\circ$

(c) (a) $\frac{1}{T} |G_{h1}(j\omega)| = [1 + (\frac{2\pi\omega}{\omega_s})^2]^{1/2} [\frac{1}{T} G_{ho}(j\omega)]^2$

$$\frac{1}{T} |G_{h1}(j2)| = [1 + (\frac{2\pi}{6})^2]^{1/2} (0.9549)^2 = 1.3203$$

$$\angle G_{h1}(j2) = \tan^{-1}(\frac{2\pi}{6}) - (\frac{2\pi}{6}) = 46.32^\circ - 60^\circ = -13.7^\circ$$

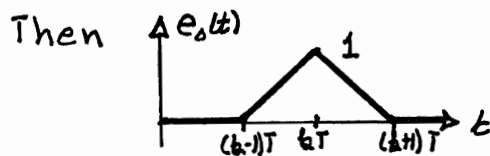
(b) $\frac{1}{T} |G_{h1}(j10)| = [1 + (\frac{20\pi}{12})^2]^{1/2} [0.1910]^2 = 0.194$

$$\angle G_{h1}(j10) = \tan^{-1}(\frac{20\pi}{12}) - (\frac{20\pi}{12}) = 79.2^\circ - 300^\circ = -220.8^\circ$$

(d) The components at $\omega=10$ are approximately equal.

The component at $\omega=10$ for the 1st order hold is approximately 30% too large, while that for the zero-order hold is approximately 10% too small. $\therefore$ the zero-order hold is better in this case.

3-19. Assume: input = $e_i(t) = \delta(t - kT)$, $\therefore E_i(s) = e^{-kTs}$



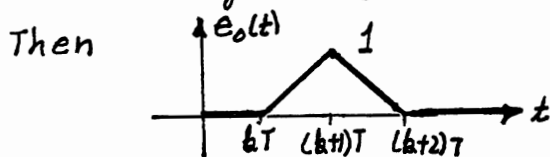
$$\therefore e_o(t) = \frac{1}{T} \{ t - (k-1)T \} u[t - (k-1)T] - \frac{2}{T} \{ t - kT \} u[t - kT] + \frac{1}{T} \{ t - (k+1)T \} u[t - (k+1)T]$$

$$\therefore E_o(s) = \frac{1}{Ts^2} e^{-(k-1)Ts} - \frac{2}{Ts^2} e^{-kTs} + \frac{1}{Ts^2} e^{-(k+1)Ts}$$

$$\therefore G_{ha}(s) = \frac{E_o(s)}{E_i(s)} = \frac{e^{-Ts}}{Ts^2} (1 - e^{-Ts})^2$$

Not physically realizable, since an advance in time is required.

3-20. Assume: input = $e_i(t) = \delta(t - kT)$, $\therefore E_i(s) = e^{-kTs}$



$$e_o(t) = \frac{1}{T} \{ t - kT \} u[t - kT] - \frac{2}{T} \{ t - (k+1)T \} u[t - (k+1)T] + \frac{1}{T} \{ t - (k+2)T \} u[t - (k+2)T]$$

$$E_o(s) = \frac{1}{Ts^2} e^{-kTs} - \frac{2}{Ts^2} e^{-(k+1)Ts} + \frac{1}{Ts^2} e^{-(k+2)Ts}$$

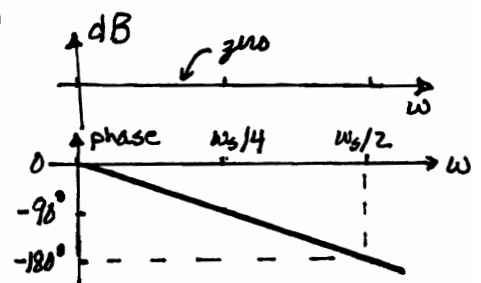
$$\therefore G_{hb}(s) = \frac{E_o(s)}{E_i(s)} = \frac{1}{Ts^2} (1 - e^{-Ts})^2$$

This data hold is physically realizable.

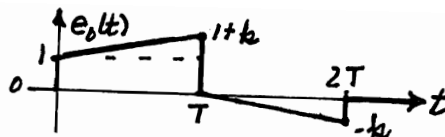
3-21. From Problems 3-19 and 3-20,

$$\frac{G_{ha}(s)}{G_{hb}(s)} = e^{-j\omega T} = \underline{1/-\omega T}$$

$$-\omega T \Big|_{\omega = \omega_0/2} = -\frac{\pi}{T} T = -\pi$$



3-22. $E_i(s) = 1$;



$$e_o(t) = u(t) + \frac{k}{T} t u(t) - (1+k) u(t-T) - \frac{2k}{T} (t-T) u(t-T) + k u(t-2T) + \frac{k}{T} (t-2T) u(t-2T)$$

$$E_o(s) = \frac{1}{s} [1 - (1+k)e^{-Ts} + k e^{-2Ts}] + \frac{k}{Ts^2} (1 - 2e^{-Ts} + e^{-2Ts})$$

$$\therefore G_{hk}(s) = \frac{E_o(s)}{E_i(s)} = (1 - k e^{-Ts}) \left(\frac{1 - e^{-Ts}}{s} \right) + \frac{k}{T} \left(\frac{1 - e^{-Ts}}{s} \right)^2$$

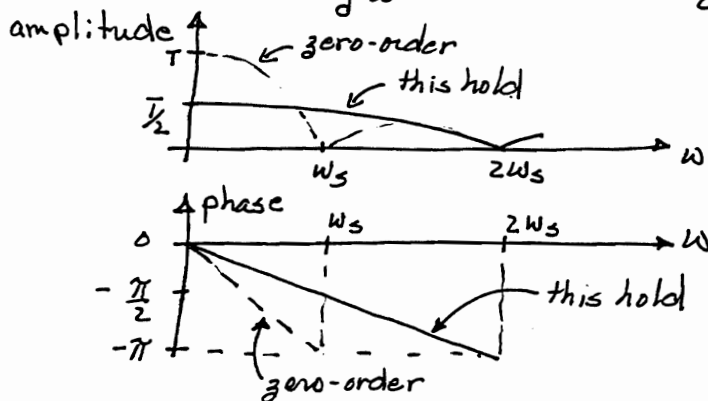
3-23. $\bar{e}(t) = e(\omega) [u(t) - u(t - \frac{T}{2})] + e(T) [u(t-T) - u(t - \frac{3T}{2})] + \dots$

$$\therefore \bar{E}(s) = e(\omega) \left[\frac{1}{s} - \frac{e^{-\frac{T}{2}s}}{s} \right] + e(T) \left[\frac{e^{-Ts}}{s} - \frac{e^{-\frac{3T}{2}s}}{s} \right] + \dots$$

$$= \left[\frac{1 - e^{-\frac{T}{2}s}}{s} \right] [e(\omega) + e(T) e^{-Ts} + \dots]$$

$$\therefore G_h(s) = \frac{1 - e^{-\frac{T}{2}s}}{s}$$

$$G_h(j\omega) = \frac{1 - e^{-j\frac{\omega T}{2}}}{j\omega} e^{j\frac{\omega T}{4}} e^{-j\frac{\omega T}{4}} = \frac{T}{2} \left[\frac{\sin(\frac{\pi\omega}{2\omega_s})}{\frac{\pi\omega}{2\omega_s}} \right] e^{-j\frac{\pi\omega}{2\omega_s}}$$



Amplitude characteristic of zero-order much better.
Phase characteristic of zero-order has more lag.

3-24. $V_o = V_{fs} [A_1 2^{-1} + A_2 2^{-2} + A_3 2^{-3} + A_4 2^{-4}]$; $A_i = 0 \text{ or } 1$

(a) $V_o = 10 [2^{-1} + 2^{-2} + 2^{-3} + 2^{-4}] = 10(0.9375) = \underline{9.375 V}$

(b) $\Delta V = 10(2^{-4}) = \underline{0.625 V}$

(c) $V_o = 10 [A_1 2^{-1} + \dots + A_4 2^{-4}]$

$\therefore V_o = \underline{0, 1.25, 2.50, 3.75, 5.00, 6.25, 7.50, 8.75}$

$$3-24 (d) V_o = 10[A_1 2^{-1} + \dots + A_n 2^{-n}]$$

$$\therefore 10(2^{-n}) = 0.005 \text{ V}$$

$$2^{-n} = 0.0005 = \frac{1}{2000} ; \therefore 2^{10} = 1024 \text{ and } 2^{11} = 2048$$

$\therefore$ 11 bits required

3-25. (a) Maximum error is a change in the least significant bit.

$$\therefore (\text{error})_{\max} = 10\left(\frac{1}{256}\right) = 0.03906 \text{ V} \approx 0.04 \text{ V}$$

(b) In Fig. 3-22, $V_x < V_R$.

$$\text{In (3-41), } V_R = 10\left[\frac{1}{2^6}\right] = 0.15625$$

The next lower value is, in binary, 0000 0011.

$$\therefore V_R = 10\left[\frac{1}{2^7} + \frac{1}{2^8}\right] = 10\left[\frac{3}{256}\right] = 0.1171875$$

$$\therefore 0.1171875 < V_x < 0.15625$$

$$3-26. n=12, f_c = 5 \text{ MHz}, \therefore T = \frac{1}{5 \times 10^6} = \underline{0.2 \text{ } \mu\text{sec}}$$

$$(a) 2^{12} = 4,096; \text{ time} = (4096)(0.2 \text{ } \mu\text{sec}) = \underline{819.2 \text{ } \mu\text{sec}}$$

$$(b) \text{ time} = (2^{12} + 2^{12})(0.2 \text{ } \mu\text{sec}) = \underline{1.6384 \text{ msec}}$$

$$(c) \text{ time} = \frac{n}{f_c} = \frac{12}{5 \times 10^6} = \underline{2.4 \text{ } \mu\text{sec}}$$

$$3-27. (1) 1000 \Rightarrow 5\left(\frac{1}{2}\right) = 2.5 \text{ V} \quad \therefore 1000 \Rightarrow \underline{2.5 \text{ V}}$$

$$(2) 1100 \Rightarrow 5\left(\frac{3}{4}\right) = 3.75 \text{ V} \quad \therefore 1000 \Rightarrow \underline{2.5 \text{ V}}$$

$$(3) 1010 \Rightarrow (5)\left(\frac{5}{8}\right) = 3.125 \text{ V} \quad \therefore 1010 \Rightarrow \underline{3.125 \text{ V}}$$

$$(4) 1011 \Rightarrow (5)\left(\frac{11}{16}\right) = 3.438 \text{ V} \quad \therefore 1011 \Rightarrow \underline{3.438 \text{ V}}$$

$$3-28. \text{ For the 8-bit converter, } V_R = V_R \sum_{i=1}^8 A_i 2^{-i}$$

$$\text{for } N = (1000 0000)_2 \Rightarrow V_R = 10\left[\frac{(1000 0000)_2}{2^8}\right] = 10\left[\frac{128}{256}\right] = 5.00 \text{ V}$$

$$\text{for } N = (1000 0000)_2 \Rightarrow V_R = 10\left[\frac{129}{256}\right] = 5.03906 \text{ V}$$

$$(a) V_R > V_x \therefore N = (1000 0001)_2 \Rightarrow N_D = 5.03906 \text{ V}$$

- 3-28. (b) $V_R > V_x$, $v_o = 5.03906 \text{ V}$ } either
 $V_R < V_x$, $v_o = 5.00 \text{ V}$ }
 (c) $V_R < V_x$, $v_o = 5.00 \text{ V}$
 (d) $V_R < V_x$, $v_o = 5.00 \text{ V}$
 (e) Parallel converter has roundoff -5.01 is closer to 5.00 than to 5.03906, $\therefore v_o = 5.00 \text{ V}$

- 3-29. (a) $N_{A/D} = (1000 \ 0001)_2 \rightarrow (1000 \ 0001 \ 0000)_2$ to D/A
 $v_o = 10 \left[\frac{(1000 \ 0001 \ 0000)_2}{2^{12}} \right] = 10 \left[\frac{(1000 \ 0001)_2}{2^8} \right] = 5.03906 \text{ V}$
 $\therefore$ all answers are same as in 3-28.

- 3-30. (a) $N_{A/D} = (1000 \ 0001)_2 \Rightarrow (0000 \ 1000 \ 0001)_2$ to D/A
 $\therefore v_o = 10 \left[\frac{(0000 \ 1000 \ 0001)_2}{2^{12}} \right] = 10 \left[\frac{129}{2^{12}} \right] = \frac{5.03906}{16} = 0.3149 \text{ V}$

These values are those of Problem 3-28 divided by 16.

- (b) 0.3149 V } either
 0.3125 V }

(c) 0.3125 V

(d) 0.3125 V

(e) 0.3125 V